

Rahul Suryakant Sakhare, Ph.D., P.E.

Curriculum Vitae

Office Address:

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EDUCATION

- **Doctor of Philosophy (Ph.D.)** April 2023
Purdue University at West Lafayette
Civil Engineering — Transportation & Infrastructure Systems (TIS)
- **Master of Technology (M.Tech.)** May 2018
Indian Institute of Technology (IIT) Madras, India
Civil Engineering (Transportation Engineering)
- **Bachelor of Technology (B.Tech.)** May 2017
Indian Institute of Technology (IIT) Madras, India
Civil Engineering (Minor: Management Studies)

PROFESSIONAL EXPERIENCE

- **Transportation Research Engineer** Aug 2023 – Present
Joint Transportation Research Program (JTRP)
Lyles School of Civil & Construction Engineering
Purdue University, West Lafayette, IN
- **Graduate Research Assistant** Jan 2020 – May 2023
Lyles School of Civil Engineering
Purdue University, West Lafayette, IN
- **Executive Consultant, Transportation Team** Jul 2018 – Nov 2019
Ernst & Young (EY) LLP, New Delhi, India
- **Visiting Undergraduate Research Student** May – Jul 2017
Purdue University, West Lafayette, IN
- **Summer Researcher — Purdue Undergraduate Research Experience (PURE)** May – Jul 2016
Lyles School of Civil Engineering, Purdue University, West Lafayette, IN

RESEARCH INTERESTS

Transportation Engineering, Connected Vehicle Data Applications, Big Data Analytics, Connected and Autonomous Vehicles, Traffic Safety, Work Zone Safety, Applied Research, Traffic Shock Wave Theory, Scalable Performance Measures for Transportation Agencies.

PROFESSIONAL REGISTRATION

- **Professional Engineer (PE), Indiana License No. PE12600619**
- **Google Cloud Certified Generative AI Leader**

SCIENTIFIC PUBLICATIONS

Google Scholar citations: 476 (June 30, 2026); h-index 13 and i10-index 18

Theses

- **PhD Dissertation:** Integrating Connected Vehicle Data for Operational Decision Making (<https://doi.org/10.25394/PGS.22684822.v1>) Purdue University, April 2023.
Committee: Dr. Darcy Bullock, Dr. Samuel Labi, Dr. Konstantina Gkritza, Dr. James Krogmeier
- **Master's Thesis:** Reliable Corridor Level Travel Time Estimation using Probe Vehicle Data. Indian Institute of Technology (IIT) Madras, Chennai, May 2018.
Committee: Dr. V. Lelitha Devi, Dr. Gitakrishnan Ramadurai, Dr. Atul Narayan, Dr. A. Veeraragavan

Monographs & Books

- **Sakhare, R. S.,** Desai, J., Mathew, J. K., McGregor, J., Kachler, M., & Bullock, D. M. (2024). Measuring and visualizing freeway traffic conditions: Using connected vehicle data (Joint Transportation Research Program No. TPF-5(514)). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317751> (444 downloads as of Jun. 10, 2026)

Peer-Reviewed Journal Publications

1. **Sakhare, R.S.,** Desai, J., Overall, M., Mukai, J., Pava, J., McGregor, J., & Bullock, D.M. (2026). Work Zone Performance Measures Derived from Connected Vehicle Data for Safety and Mobility Assessment. *Future Transportation*, 6(1), 12. <https://doi.org/10.3390/futuretransp6010012>
2. Desai J.C., Gartner, C.M., **Sakhare, R.S.,** Cox E. & Bullock, D.M. (2026). Evaluating Safety Benefits of Ramp Metering by Leveraging Connected Vehicle Data. *IEEE Open Journal of Intelligent Transportation Systems*, 7, 190-199. <https://doi.org/10.1109/OJITS.2025.3650100>
3. Mukai, J., Desai, J., **Sakhare, R.S.,** Sturdevant, N. and Bullock, D. (2026) Quantifying Hard Breaking Events Occurring Prior to Secondary Crashes on Indiana Interstates. *Journal of Transportation Technologies*, 16, 150-165. <https://doi.org/10.4236/jtts.2026.161010>
4. Overall, M., Mukai, J., **Sakhare, R.S.,** Desai, J., Horton, D. and Bullock, D.M. (2026). Applications of Commercial Truck Dash Cameras for Work Zone Inspection and Monitoring. *Journal of Transportation Technologies*, 16, 95-114. <https://doi.org/10.4236/jtts.2026.161006>
5. Gartner, C.M., **Sakhare, R.S.,** Desai J.C., Sturdevant J. & Bullock, D.M. (2025). A Scalable Data Model for Signalized Intersection Performance Measures. *IEEE Access*, 13, 197851-197863. <https://doi.org/10.1109/ACCESS.2025.3635007>
6. Desai, J., **Sakhare, R. S.,** McGregor, J., Snyder, B., & Bullock, D. M. (2025). Evaluating Impact of Automated Work Site Speed Enforcement Using Connected Vehicle Data. *Applied Sciences*, 15(22), 12140. <https://doi.org/10.3390/app152212140>
7. Mathew, J., Desai, J., **Sakhare, R. S.,** Hunter, J. and Bullock, D. (2025) Spatiotemporal Analysis of Pavement Roughness Using Connected Vehicle Data for Asset Management. *Journal of Transportation Technologies*, 15, 1-16. <https://doi.org/10.4236/jtts.2025.151001>
8. Overall, M. W., Mukai, J., **Sakhare, R. S.,** Desai, J., Lowther, H., & Bullock, D. M. (2025). Measuring the Impact of Recovery Resource Delay on Traffic Incident Management Clearance Times. *Future Transportation*, 5(4), 171. <https://doi.org/10.3390/futuretransp5040171>
9. Driscoll, T., **Sakhare, R.S.,** Desai, J. and Bullock, D.M. (2025) Utilizing Connected Vehicle Data to Identify Impacts of Congestion on Adjacent Agencies. *Journal of Transportation Technologies*, 15, 522-541. <https://doi.org/10.4236/jtts.2025.154024>
10. **Sakhare, R.S.,** Desai, J., Mathew, J.K. and Bullock, D.M. (2024). Assessing the Interstate Coverage of Commercial Trucks Capable of Providing Roadway Imagery via On-Vehicle Dash Camera in the United States. *IEEE Access*, vol. 12, pp. 173517-173529. <https://doi.org/10.1109/ACCESS.2024.3503368>
11. **Sakhare, R.S.,** Desai, J., Mathew, J. and Bullock, D.M. (2024). Impact of 2024 Solar Eclipse on National Traffic Mobility using Connected Vehicle Data and Images. *Transportation Research Interdisciplinary Perspectives*, 27. <https://doi.org/10.1016/j.trip.2024.101225>
12. Brinster, G. L., Desai, J., **Sakhare, R. S.,** Mathew, J. K., & Bullock, D. M. (2024). Expanding on methodologies for analyzing truck speeds in states with differential speed limits. *Safety*, 10(4), 99. <https://doi.org/10.3390/safety10040099>
13. Vajpayee, V., Saldivar-Carranza, E. D., **Sakhare, R. S.,** & Bullock, D. M. (2024). Large Scale Evaluation of Normalized Hard-Braking Events Derived from Connected Vehicle Trajectory Data at Signalized Intersections, Roundabouts, and All-Way Stops. *Future Transportation*, 4(3), 968-984. <https://doi.org/10.3390/futuretransp4030046>

14. Brinster, G.L., Desai, J., Overall, M.W., Gartner, C., **Sakhare, R.S.**, Mathew, J.K., Evans, N. and Bullock, D. (2024) Evaluating the Robustness of MDSS Maintenance Forecasts Using Connected Vehicle Data. *Journal of Transportation Technologies*, 14, 549-569. <https://doi.org/10.4236/jtts.2024.144030>
15. Saldivar-Carranza, E. D., **Sakhare, R. S.**, Desai, J., Mathew, J. K., Sivakumar, A. J., Mukai, J., & Bullock, D. M. (2024). Impact of Privacy Filters and Fleet Changes on Connected Vehicle Trajectory Datasets for Intersection and Freeway Use Cases. *Smart Cities*, 7(5), 2366-2391. <https://doi.org/10.3390/smartcities7050093>
16. Desai, J., Saldivar-Carranza, E. D., **Sakhare, R. S.**, Mathew, J. K., & Bullock, D. M. (2024). Impact of In-Cab Alerts on Connected Truck Speed Reductions in Indiana. *Vehicles*, 6(4), 1857-1871. <https://doi.org/10.3390/vehicles6040090>
17. **Sakhare, R. S.**, Desai, J., Saldivar, E. D., & Bullock, D. M. (2023). Methodology for Monitoring Border Crossing Delays with Connected Vehicle Data: United States and Mexico Land Crossings Case Study. *Future Transportation*, 4(1), 107-129. <https://doi.org/10.3390/futuretransp4010007>
18. **Sakhare, R. S.**, Li, H., & Bullock, D. M. (2023). Methodology for the Identification of Shock Wave Type and Speed in a Traffic Stream Using Connected Vehicle Data. *Future Transportation*, 3(4), 1147-1174. <https://doi.org/10.3390/futuretransp3040063>
19. **Sakhare, R. S.**, Zhang, Y., Li, H., & Bullock, D. M. (2023). Impact of Rain Intensity on Interstate Traffic Speeds Using Connected Vehicle Data. *Vehicles*, 5(1), 133-155. <https://doi.org/10.3390/vehicles5010009>
20. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2023). Analysis of Connected Vehicle Data to Quantify National Mobility Impacts of Winter Storms for Decision Makers and Media Reports. *Future Transportation*, 3(4), 1292-1309. <https://doi.org/10.3390/futuretransp3040071>
21. Desai, J., Mathew, J., Li, H., Mukai, J., **Sakhare, R. S.**, & Bullock, D. M. (2023). Correlating Connected Vehicle Estimated Deceleration Events with Crash Incidents near Interstate Interchanges. *Journal of Transportation Technologies*, 13, 674-688. <https://doi.org/10.4236/jtts.2023.134031>
22. **Sakhare, R. S.**, Hunter, M., Mukai, J., Li, H., & Bullock, D. M. (2022). Truck and Passenger Car Connected Vehicle Penetration on Indiana Roadways. *Journal of Transportation Technologies*, 12, 578-599. <https://doi.org/10.4236/jtts.2022.124034>
23. **Sakhare, R. S.**, Desai, J., Li, H., Kachler, M. A., Bullock, D. M. (2022). Methodology for Monitoring Work Zones Traffic Operations Using Connected Vehicle Data. *Safety*, 8(2), 41. **(Editor's Choice Award)** <https://doi.org/10.3390/safety8020041>
24. Mahlberg, J. A., Desai, J., Li, H., **Sakhare, R. S.**, Wells, T., Bullock, D. M. (2022). Calibration and Confidence in Snowplow Fleet Operations and Fleet Telematics. *Journal of Transportation Technologies*, 12(4), 696-710. <https://doi.org/10.4236/jtts.2022.124040>
25. **Sakhare, R. S.**, Desai, J. C., Mahlberg, J., Mathew, J. K., Kim, W., Li, H., McGregor, D., & Bullock, D. M. (2021). Evaluation of the Impact of Queue Trucks with Navigation Alerts Using Connected Vehicle Data. *Journal of Transportation Technologies*, 11(4), 561-576. <https://doi.org/10.4236/jtts.2021.114035> **(Recognized by ITS Midwest as the 2021 Project of the Year)**
26. **Sakhare, R. S.**, Desai, J. C., Mathew, J. K., McGregor, J. D., & Bullock, D. M. (2021). Evaluation of the impact of presence lighting and digital speed limit trailers on interstate speeds in Indiana work zones. *Journal of transportation technologies*, 11(2), 157-167. <https://doi.org/10.4236/jtts.2021.112010>
27. Mathew, J. K., Desai, J. C., **Sakhare, R. S.**, Kim, W., Li, H., & Bullock, D. M. (2021). Big data applications for managing roadways. *Institute of Transportation Engineers. ITE Journal*, 91(2), 28-35.
28. Mahlberg, J. A., **Sakhare, R. S.**, Li, H., Mathew, J. K., Bullock, D. M., & Surnilla, G. C. (2021). Prioritizing Roadway Pavement Marking Maintenance Using Lane Keep Assist Sensor Data. *Sensors*, 21(18), 6014. <https://doi.org/10.3390/s21186014>
29. Desai, J., Mahlberg, J., Kim, W., **Sakhare, R. S.**, Li, H., McGuffey, J., & Bullock, D. M. (2021). Leveraging telematics for winter operations performance measures and tactical adjustment. *Journal of Transportation Technologies*, 11(4), 611-627. <https://doi.org/10.4236/jtts.2021.114038>
30. **Sakhare, R. S.** & Vanajakshi, L. (2020). Reliable Corridor Level Travel Time Estimation Using Probe Vehicle Data. *Transportation Letters*, 12:8, 570-579. <https://doi.org/10.1080/19427867.2019.1671041>
31. **Sakhare, R. S.**, Dev, L. K., Prasad, K. E., & Robinson, R. G. (2016). A Double Acting Piston Based Automatic Volume Change Apparatus. *Geotechnical Testing Journal*, 39(6), 899-905. <http://doi.org/10.1520/GTJ20150296> **(IGS Best Paper Award)**

Key Project / Technical Reports

1. Thompson, A., Desai, J. C., **Sakhare, R. S.**, & Bullock, D. M. (2026). Research support to INDOT on systemwide asset condition assessment using connected vehicle data (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2026/04). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318612>
2. **Sakhare, R. S.**, Desai, J., Gartner, C., Mukai, J., Sturdevant, N., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D. M. (2025). Connected vehicle trajectory data to screen network for hard-braking and hard-acceleration events (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/30). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318579>
3. Desai, J., **Sakhare, R. S.**, McGregor, J., Sturdevant, J., & Bullock, D. M. (2025). Worksite speed control system (WSCS) (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/33). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318595>
4. Overall, M. W., Mukai, J., Desai, J., **Sakhare, R. S.**, Kachler, M., McGregor, J., & Bullock, D. M. (2025). Continued deployment of Indiana work zone analytics (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/31). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318581>
5. Desai, J. C., Gartner, C., **Sakhare, R. S.**, Cox, E. D., Sturdevant, J., & Bullock, D. M. (2025). Research support to INDOT on I-465 Southeast variable speed limit and ramp meter project (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/32). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318592>
6. Overall, M. W., Mukai, J., **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). Case studies documenting lessons learned and identifying opportunities for maintenance of traffic (MOT) improvements (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/29). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318578>
7. Desai, J., **Sakhare, R. S.**, Mathew, J. K., Sturdevant, N., Cox, E., & Bullock, D. M. (2025). Communication of fixed and mobile warnings to commercial trucks using in-cab notification (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/24). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284318232>
8. Overall, M. W., Mukai, J., **Sakhare, R. S.**, Desai, J., Lowther, H., Cox, E., Sturdevant, J., & Bullock, D. M. (2025). Development of incident management performance measures database and supporting training material (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/21). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317907>
9. **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). Crawfordsville District interstate speed profiles: April 2020–March 2025. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317852> (1,888 downloads as of 2/26/2026)
10. **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). *Indiana interstate speed profiles 2020–2024*. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317843>
11. Gayen, S., Saldivar-Carranza, E. D., **Sakhare, R. S.**, Desai, J., Mathew, J. K., Shellhamer, N., Ruble, T., McCoy, D., Gallagher, J., & Bullock, D. M. (2024). Statewide screening of signalized intersections for capacity improvements (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2024/20). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317755> (128 downloads as of 10/15/2025)
12. Brinster, G. L., Desai, J., Overall, M. W., Gartner, C., **Sakhare, R. S.**, Mathew, J. K., Evans, N., & Bullock, D. M. (2024). Evaluating the robustness of MDSS forecast and compliance with recommendations (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2024/31). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317805> (49 downloads as of 10/15/2025)
13. **Sakhare, R. S.**, Desai, J., Woker, W., Li, H., Mathew, J., Mahlberg, J., Saldivar-Carranza, E. D., Horton, D., & Bullock, D. M. (2023). Connected vehicle centric dashboards for TMC of the future (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/17). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317642> (248 downloads as of 10/15/2025)
14. Desai, J., **Sakhare, R. S.**, Mahlberg, J., Mathew, J. K., Li, H., & Bullock, D. M. (2023). Implementation of enhanced probe data (CANBUS) for tactical workzone and winter operations management (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/18). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317643> (134 downloads as of 10/15/2025)
15. Mahlberg, J. A., Desai, J., Malackowski, H., Li, H., Mathew, J. K., **Sakhare, R. S.**, Wells, T., & Bullock, D. M. (2023). Crowdsourcing/winter operations dashboard upgrade (Joint Transportation Research Program Publication

- No. FHWA/IN/JTRP-2023/21). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317652> (166 downloads as of 10/15/2025)
16. **Sakhare, R. S.**, Mahlberg, J., Mathew, J. K., Desai, J., Li, H., Bullock, D. M. (2022). Impacts to traffic behavior from queue warning truck: Current pilot project (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2022/26). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317448> (284 downloads as of 10/15/2025)
 17. Mathew, J. K., Li, H., Desai, J., **Sakhare, R. S.**, Saldivar-Carranza, E., Hunter, M., Scholer, B., & Bullock, D. M. (2022). Integration of probe data tools into TMC operations (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2022/01). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317363> (348 downloads as of 10/15/2025)
 18. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2022). National mobility analysis for all interstate routes in the United States: December 2022. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317591> (284 downloads as of 10/15/2025)
 19. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2022). National mobility analysis for all interstate routes in the United States: August 2022. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317585> (333 downloads as of 10/15/2025)
 20. **Sakhare, R. S.**, Li, H., Mathew, J. K., Desai, J., Horton, D., & Bullock, D. M. (2022). Indiana interstate speed profiles 2018–2022. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317589> (153 downloads as of 10/15/2025)
 21. **Sakhare, R. S.**, Desai, J., Mathew, J. K., Kim, W., Mahlberg, J., Li, H., & Bullock, D. M. (2021). Evaluating the impact of vehicle digital communication alerts on vehicles (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2021/19). West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284317324> (409 downloads as of 10/15/2025)
 22. Day, C. M., McNamara, M. L., Li, H., **Sakhare, R. S.**, Desai, J., Cox, E. D., Horton, D. K., & Bullock, D. M. (2016). 2015 Indiana mobility report and performance measure dashboards. West Lafayette, IN: Purdue University. <https://doi.org/10.5703/1288284316352> (720 downloads as of 10/15/2025)

Peer-Reviewed Conference & Workshop Presentations

1. **Sakhare, R.S.**, Desai, J., Overall, M., Mukai, J., Pava, J., McGregor, J., & Bullock, D. M. (2026, January). Improving Vehicle and Worker Safety in Dynamic Work Zones – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02289). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24853>
2. Desai, J., Gartner, C., **Sakhare, R.S.**, Cox, E., & Bullock, D. M. (2026, January). Improving Vehicle and Worker Safety in Dynamic Work Zones – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02345). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24853>
3. **Sakhare, R.S.**, Desai, J., Gartner, C., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D. M. (2026, January). Travel Data and Networks: Collection, Modeling, and Mobility Insights – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-01332). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24601>
4. Gartner, C., **Sakhare, R.S.**, Desai, J., Sturdevant, J., & Bullock, D. M. (2026, January). Advanced Modeling, Optimization, and Safety for Signalized Intersections – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02156). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24854>
5. Desai, J., **Sakhare, R.S.**, McGregor, J., Snyder, B., & Bullock, D. M. (2026, January). Data-Driven Approaches for Perception, Prediction, and Decision Making in Autonomous Vehicles – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02221). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24522>
6. Overall, M., Mukai, J., **Sakhare, R.S.**, Desai, J., Horton, D., & Bullock, D. M. (2026, January). Modeling Traffic Speed and Network Performance Under Disruptive Events – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-03560). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24845>
7. Driscoll, T., **Sakhare, R.S.**, Desai, J., & Bullock, D. M. (2026, January). Emerging Research Topic in Strategic Management and Organizational Excellence – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-03893). TRB. <https://annualmeeting.mytrb.org/OnlineProgram/Details/24765>

8. Overall, M., Mukai, J., **Sakhare, R.S.**, Desai, J., Hillary, L., & Bullock, D. M. (2026, January). Frameworks for Connected Traffic and Emergency Operations Management – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-03559). TRB.
<https://annualmeeting.mytrb.org/OnlineProgram/Details/24533>
9. Mathew, J., Desai, J., **Sakhare, R.S.**, Hunter, J., & Bullock, D. M. (2025, January). Innovative Use of Emerging Data and Criteria for Maintenance Management – Lectern Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-01326). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22915>
10. Brinster, G., Desai, J., Overall, M., Gartner, C., **Sakhare, R.S.**, Mathew, J., Evans, N., & Bullock, D. M. (2025, January). Innovative Use of Emerging Data and Criteria for Maintenance Management – Lectern Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-02443). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22915>
11. **Sakhare, R.S.**, Desai, J., Mathew, J., & Bullock, D. M. (2025, January). Statewide and National Transportation Data Research – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-03615). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22967>
12. Bullock, D. M., & **Sakhare, R.S.** (2025, January). State Department of Transportation Research: High-Value Project Winners – Poster Session. In Transportation Research Board 104th Annual Meeting (No. P25-20182). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22727>
13. Saldivar-Carranza, E., **Sakhare, R.S.**, Desai, J., Mathew, J., Sivakumar, A. J., Mukai, J. & Bullock, D. M. (2025, January). Intelligent Transportation Systems in the Era of Connected vehicles and Vehicle to Everything Communication – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-03624). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22967>
14. Desai, J., Saldivar-Carranza, E., **Sakhare, R.S.**, Mathew, J., & Bullock, D. M. (2025, January). Toward Better Driving: Innovations for Improved Driving Performance and Enhanced Safety – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-02784). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22931>
15. Desai, J., Mathew, J., Li, H., **Sakhare, R.S.**, Horton, D., & Bullock, D. M. (2024, January). Statewide and National Data and Information Systems – Poster Session. In Transportation Research Board 103rd Annual Meeting (No. TRBAM-24-04421). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/21103>
16. **Sakhare, R.S.**, Zhang, Y., Li, H., & Bullock, D. M. (2023, January). Road Weather Management Solutions for Improved Safety and Operations – Lectern Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-02713). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19287>
17. Mahlberg, J., Desai, J., Li, H., **Sakhare, R.S.**, Wells, T., & Bullock, D. M. (2023, January). Advancements Available to Maintenance Equipment Fleets Within Departments of Transportation and Other Fleet Operations – Lectern Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-01921). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19002>
18. **Sakhare, R.S.**, Hunter, M., Mukai, J., Li, H., & Bullock, D. M. (2023, January). Road Scholars: New Research in Travel Time, Speed, and Reliability Data – Poster Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-01814). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19057>
19. Desai, J., Mahlberg, J., Woosung, K., **Sakhare, R.S.**, Li, H., McGuffey, J., & Bullock, D. M. (2022, January). Advancements in Winter Maintenance – Lectern Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-02228). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17477>
20. **Sakhare, R.S.**, Desai, J., Mahlberg, J., Mathew, J., Woosung, K., Li, H., McGregor, J., & Bullock, D. M. (2022, January). Freeway and Regional Transportation Systems and Operations – Poster Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-02130). TRB.
<https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17526>
21. Mahlberg, J., **Sakhare, R.S.**, Li, H., Mathew, J., Bullock, D. M. & Surnilla, G. (2022, January). Traffic Control Devices – Poster Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-01292). TRB. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17477>
22. Desai, J., **Sakhare, R. S.**, Rogers, S., Mathew, J. K., Habib, A., & Bullock, D. (2021, September). Using Connected Vehicle Data to Evaluate Impact of Secondary Crashes on Indiana Interstates. In 2021 IEEE International Intelligent Transportation Systems Conference (ITSC) (pp. 4057-4063). IEEE.
<https://doi.org/10.1109/ITSC48978.2021.9564653>

23. **Sakhare, R. S.**, Desai, J. C., Li, H., Mathew, J. K., & Bullock, D. M. (2021, January). Predicting Weather Impact on Traffic Speeds Using the Multi-Radar Multi-Sensor System. In 101st American Meteorological Society Annual Meeting. AMS. <https://ams.confex.com/ams/101ANNUAL/meetingapp.cgi/Paper/382590>
24. **Sakhare, R. S.**, Mathew, J. K., Avr, A., Hubbard, S. M., Devi, L., & Bullock, D. M. (2018, January). Comparison of Bluetooth and Bus GPS Data for Estimating Arterial Travel Time and Trip Chaining. In Transportation Research Board 97th Annual Meeting (No. TRBAM-18-01695). TRB. <https://trid.trb.org/View/1495014>
25. Li, H., **Sakhare, R. S.**, Mathew, J. K., Mackey, J., & Bullock, D. M. (2018, January). Estimating Intersection Control Delay Using High Fidelity Commercial Probe Vehicle Trajectory Data. In Transportation Research Board 97th Annual Meeting (No. TRBAM-18-01695). TRB. <https://trid.trb.org/View/1494388>
26. McNamara, M., **Sakhare, R. S.**, Li, H., Baldwin, M., & Bullock, D. M. (2017, January). Integrating Crowdsourced Probe Vehicle Traffic Speeds into Winter Operations Performance Measures. In Transportation Research Board 96th Annual Meeting (No. TRBAM-17-00161). TRB. <https://trid.trb.org/View/1437147>

Key Technical Presentations

1. **Sakhare, R.S.**, & Taylor, R. (2026, March). Segment Mobility and Signal Performance: Leveraging CV Metrics – Lectern Session. In 112th Purdue Road School (No. 131).
2. **Sakhare, R.S.**, Thompson, A., & Sturdevant, N. (2026, March). Evaluation of Asset Identification Technology Utilizing Truck Images – Lectern Session. In 112th Purdue Road School (No. 159).
3. **Sakhare, R.S.**, & McGregor, J. (2026, March). Utilizing CV Data and Truck Imagery for Roadway Monitoring – Lectern Session. In 112th Purdue Road School (No. 159).
4. Gartner, C., & **Sakhare, R.S.** (2026, February). Connected Vehicle Data Use Cases for Practitioners – Lectern Session. In 2026 ITE Indiana Section February Tech Lunch. <https://indianaite.starchapter.com/meetinginfo.php?id=202&ts=1768590344>
5. **Sakhare, R.S.** (2026, February). Work Zone Performance Measures Derived from Connected Vehicle Data that Map into the FHWA Rule on Work Zone Safety and Mobility – Seminar. In Purdue ITE Weekly Seminar Series.
6. **Sakhare, R.S.** (2023, April). Impact of Rain Intensity on Interstate Traffic Speeds Using Connected Vehicle Data – Seminar. In Purdue ITE Weekly Seminar Series.
7. **Sakhare, R.S.**, & Hillary, L. (2023, March). Identifying Traffic Incident Management Event Sequence using CV Data, Crash Reports, and ITS Images – Lectern Session. In 109th Purdue Road School (No. 61).
8. **Sakhare, R.S.** (2023, March). Automation System Performance Issues/Truck and Car CV Penetration – Lectern Session. In 109th Purdue Road School (No. 117).
9. **Sakhare, R.S.**, & Desai, J. (2023, March). Methodology for Monitoring Work Zones Traffic Operations using CV Data: Implementation on Indiana Interstate with UAS Imagery – Lectern Session. In 109th Purdue Road School (No. 182).

FUNDED RESEARCH PROJECTS

(PI: Principal Investigator; RE: Research Engineer; GRA: Graduate Research Assistant)

- **FHWA** — TPF-5(514): Work Zone Analytics (RE, 2023-2026)
- **FHWA** — TPF-5(519): Expansion: Enhanced Traffic Signal Performance Measures (RE, 2023-2026)
- **INDOT** — SPR-5005: Evaluation of Asset Identification Technology (Co-PI, 2025-2027)
- **INDOT** — SPR-5021: Traffic Signal Freight Prioritization via Vehicle to Infrastructure (V2I) Communications (Co-PI, 2026-2027)
- **INDOT** — SPR-5020: Identifying Locations with Abnormally High Wrong Way Driving or Interstate U-Turns (Co-PI, 2025-2027)
- **INDOT** — SPR-4928: Connected Vehicle Trajectory Data to Screen Network for Hard Braking and Hard Acceleration Events (Co-PI, 2024-2025)
- **INDOT** — SPR-4933: Worksite Speed Control System (WSCS) (RE, 2024-2025)
- **INDOT** — SPR-4907: Systemwide Asset Condition Assessment using Connected Vehicle Data (RE, 2024-2025)
- **INDOT** — SPR-4857: Statewide Screening of Signalized Intersections for Capacity Improvements (RE, 2023-2024)
- **INDOT** — SPR-4854: MOT for MOT (RE, 2023-2025)

- **INDOT** — SPR-4851: Development of Incident Management Performance Measure Database and IN Time Training Material (RE, 2023-2025)
- **INDOT** — SPR-4850: Continued Deployment of Indiana Work Zone Analytics (RE, 2023-2025)
- **INDOT** — SPR-4803: Communication of Fixed and Mobile Warning to Commercial Trucks Using in Cab Notification (RE, 2023-2025)
- **INDOT** — SPR-4704: Evaluating the Robustness of MDSS Forecast and Compliance with Recommendations (RE, 2022-2024)
- **INDOT** — SPR-4639: Connected Vehicle Centric Dashboards for TMC of the Future (GRA, 2021-2023)
- **INDOT** — SPR-4636: Research Support to INDOT on I-465 Southeast Variable Speed Limit and Ramp Meter Project (RE, 2021-2025)
- **INDOT** — SPR-4603: Crowdsourcing/Winter Operations Dashboard Upgrade (GRA, 2021-2023)
- **INDOT** — SPR-4536: Implementation of Enhanced Probe Data (CANBUS) for Tactical Workzone and Winter Operations Management (GRA, 2020-2023)
- **CCAT** — Methodologies for Monitoring Work Zone Performance Measures across State Border Highways using Connected Vehicle Data (RE, 2023-2025)
- **INDOT** — SPR-4600: Impacts to Traffic Behavior from Queue Warning Truck – Current Pilot Project (GRA, 2021-2022)
- **INDOT** — SPR-4524: Evaluation of the Impact of On Vehicle Digital Communication Alerts to Improve INDOT and Motorist Safety (GRA, 2020-2022)
- **INDOT** — SPR-4451: Integration of Probe Data Tools into TMC Operations (GRA, 2020-2022)

HONORS, AWARDS & ACHIEVEMENTS

- **2024** — Google Cloud Research Innovator part of the [fourth cohort](#)
- **2023** — Edward J. Cox Memorial Scholarship, Indiana ITE Section
- **2020** — International Collegiate Traffic Bowl Championship, ITE International
- **2016** — Gold Medal in Civil Engineering, National Design and Research Forum, Institute of Engineers, India
- **2016** — Mr. H. C. Verma Diamond Jubilee Awards, Indian Geotechnical Society (IGS), India
- **2014** — Village's Choice Award, Faculty of Engineering, Chulalongkorn University, Thailand
- **2014** — Represented India at the Engineering Case Competition (ECC) in Thailand
- **2018** — Exemplary and All-round Best Performance in Dual Degree Program, Civil Engineering, Indian Institute of Technology (IIT) Madras, India
- **2021** — ITS Midwest Project of the Year Award Winner
- **2022-23** — Best Student Speaker Award, Purdue ITE
- **2020** — Best Poster Presentation Sigma Xi Scientific Society, Purdue University
- **2021 & 2022** — Elevator Pitch Event Winner, Purdue ITE
- **2021** — Student Team Design Competition Winner, Great Lakes District ITE
- **2020-2023** — Christopher B. & Susan S. Burke Graduate Program Research Assistantship, Purdue University
- **2016** — Purdue Undergraduate Research Experience (PURE) Program Scholar; Among the top 3 Civil Engineering students chosen across India
- **2017** — Invited Visiting Undergraduate Student at Purdue University
- **2011** — Ranked 4th in the 39th Mathematical Olympiad at Junior Inter Level, Andhra Pradesh, India
- **2008** — All India Open Mathematics Scholarship recipient; Ranked 41st across India
- **2018** — Winner at CEA Technical Events (Case Study, Prabandha, Bon Auto and Aquanomics), IIT Madras, Chennai, India
- **2013** — Ranked top 0.2 percentile (ranked 2,445) of 1.45 million students at IIT Joint Entrance Exam; considered amongst world's toughest examinations
- **2009** — Maharashtra Talent Search Examination (MTSE) Scholarship Awardee
- **2009** — Maharashtra State Certificate in Information Technology (MS-CIT)
- **2011** — Scored 100% in Secondary School Certificate Examination (10th Standard)

- **2016** — Represented India (Team of four) at Engineering Case Competition in Thailand organized by the Faculty of Chulalongkorn University
- **2015** — Winner of Smart City Challenge Vellore-Engineers for Life Program
- **2015** — Successfully Completed the Engineers for Life Program, AidVision Initiative

Non-Academic

- **2010** — Competed at State Level Archery Competitions, Maharashtra, India
- **2022** — National Collegiate Cricket Association's (NCCA) Midwest Zone Championship Winner (Captain of the Purdue Cricket team)
- **2022** — Midwest Cricket Tournament Runner-up (more than 60 participating teams)
- **2022 & 2023** — Purdue Indoor Cricket League Winner (Captain of the team in 2023)
- **2018** — Winner in CEA Cricket Tennis Ball Tournament (runner up in 2017)
- **2017** — Fourth in Men's Cricket, IITM Institute Gymkhana Intramural Competitions

ACADEMIC SERVICE

- Transportation Research Board (TRB) **Committee Member** ACF13 - Standing Committee on Limited Access Roadway Operations *(August 2025 – current)*
- Purdue Road School **Planning Committee Member** Urban Planning track (2025) Electric & Connected Vehicles track (2026) Maintenance & Operations track (2026) Safety track (2026) *(October 2024 – Present)*
- **Session Moderator** Safety track (2023) Project Management/Planning track (2024) Project Management/Planning track (2025) *(2023 – 2025)*
- Institute of Transportation Engineers (ITE) **Executive Board Member – Purdue Chapter** Received ITE Student Chapter Award from ITE International Received *Outstanding Student Chapter from ITE Great Lakes District (2021 – 2022)*
- Judge for 2024 Spring Undergraduate Research Conference *(2024)*
- Head Member at International & Alumni Relations Student Council, IIT Madras *(2016)*
- Head Member at Civil Engineering Association (CEA), IIT Madras *(2015)*
- Ambience & Design Super-Coordinator at Shaastra, Technical Event, IIT Madras *(2015)*
- Student Mentor, MITR Office of the Dean Students, IIT Madras *(2015)*
- Events & Projects Coordinator at Sustainability Network, IIT Madras *(2014)*
- Volunteer for Annual Terry Fox Run, The Terry Fox Foundation, Chennai *(2016)*
- Captained Purdue Cricket Team at NCCA Finals Tournament *(2022)*

TEACHING & MENTORING EXPERIENCE

- **Instructor** for Mobility Course and Workshop SPARC (Scheme for Promotion of Academic and Research Collaboration, Government of Indian Initiative) - Developed course material and taught 3 session modules *(2021)*
- **Teaching Assistant** for CE2080 Surveying course Department of Civil Engineering, IIT Madras *(2017-2018)*

SCHOLARLY PROFILES

- **Google Scholar:** <https://scholar.google.com/citations?user=4crwCDoAAAAJ&hl=en>
- **Research Gate:** <https://www.researchgate.net/profile/Rahul-Suryakant-Sakhare>
- **ORCID:** <https://orcid.org/0000-0001-7843-5707>
- **LinkedIn:** <https://www.linkedin.com/in/rahulsakhare/>